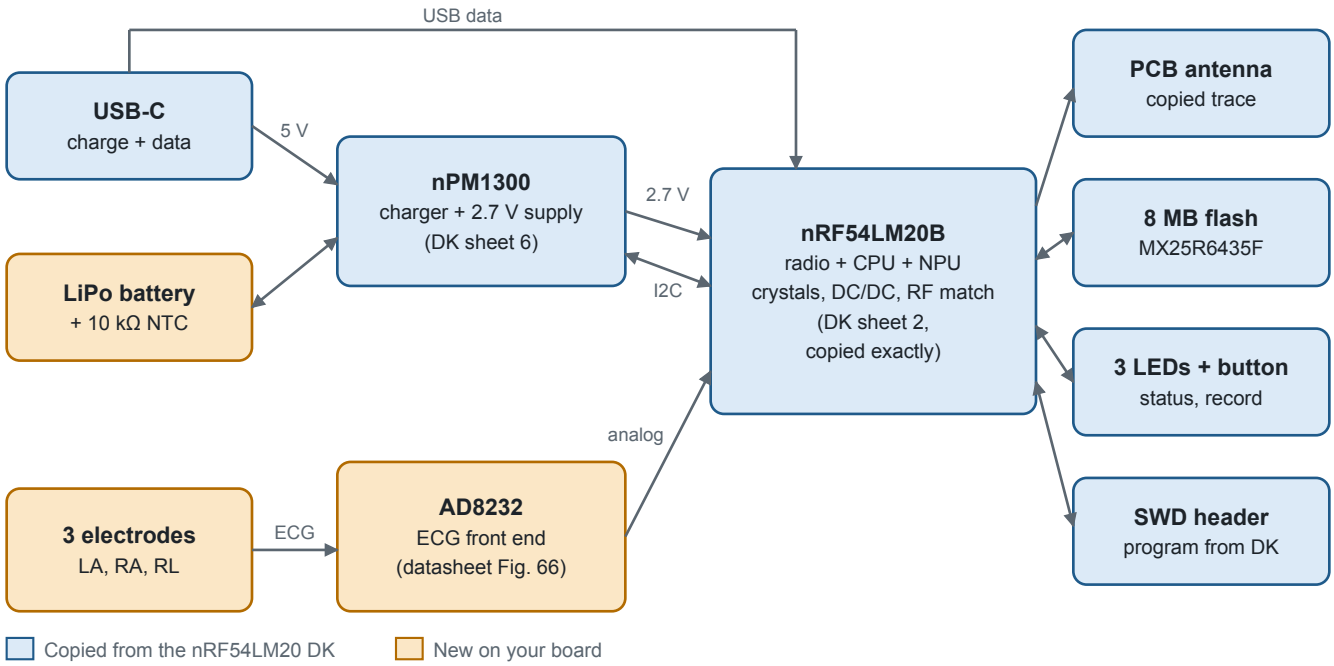
LeoPatch Rev 1 PCB Plan

A custom board built from the Nordic nRF54LM20 DK, with an AD8232 ECG front end and a rechargeable battery

25 October 2026 · Based on the nRF54LM20 DK hardware files v0.7.0 (PCA10184) and the datasheets listed in Section 8 · For testing on adult team members only, not infants

1. What goes on the board

The board is Nordic's nRF54LM20 DK with the development-only parts removed and three things added: an AD8232 ECG front end, electrode connectors, and a rechargeable battery. Everything around the nRF54LM20B chip is copied unchanged from Nordic's design, which is what Nordic recommends: *"use the PCB layouts and component values provided by Nordic Semiconductor"* (nRF54LM20 datasheet, section 10.3). Because it is the same chip with the same supporting parts, the firmware and model that already run on your DK will run on this board after a small pin-map change.



Block	Main part (manufacturer part number)	Comes from	Change from the DK
Radio core	nRF54LM20B-PAAA; 32 MHz and 32.768 kHz crystals; DC/DC inductor; RF tuning parts; test connector; PCB antenna	DK sheet 2	None. Copy schematic and layout exactly
Power and charging	nPM1300-QEAA; 1-cell LiPo; 10 kΩ NTC thermistor	DK sheet 6	Battery added; supply raised to 2.7 V; controlled by the nRF54 instead of the debugger
USB-C	Bellwether 80259-1021; Nexperia PRTR5V0U2F (ESD); Murata DLM0NSM900HY2D (choke)	DK sheet 8, "nRF54LM20 USB" port	One port only; its power and CC pins go to the nPM1300
Storage	Macronix MX25R6435FZNIL0, 8 MB flash	DK sheet 5	Wired straight to the chip (switches removed)
ECG front end	Analog Devices AD8232ACPZ-R7	AD8232 datasheet, Fig. 66	New

Block	Main part (manufacturer part number)	Comes from	Change from the DK
Electrodes	3 connectors: LA, RA, RL	AD8232 datasheet	New
Status LEDs	3 LEDs, each with a Rohm RV2C010UNT2L switch	DK sheet 5	3 of the DK's 4 LED circuits; green, amber and red. Covers recording, done, and memory nearly full
Buttons	Record/upload button; reset button	DK sheet 5	1 of 4 user buttons kept
Programming	2×5 1.27 mm SWD header (WCON 3132-10MS0BK00R5)	DK sheet 8, P11 "Debug IN"	Fitted (empty on the DK)
Circuit board	8 copper layers, 1.57 mm FR-4, ENIG, filled and capped vias	DK laminate spec 400861	Same

Decisions made so far

- Copy the DK rather than use a pre-made module: closest to Nordic's design, and it is the board the firmware already runs on.
- AD8232 ECG front end with 3 electrodes.
- Rechargeable LiPo, charged over USB-C by the DK's nPM1300.
- One 2.7 V supply for the nRF54, flash and AD8232.
- Designed in KiCad 9, starting from an import of Nordic's Altium files.

2. Choices: pros and cons

2.1 Radio core: copy the DK, or use a pre-made module

	Copy the DK (chosen)	Pre-made module, e.g. Fanstel BM20BF
Pros	Identical to the board your firmware and model already run on. Nordic's own tested layout and parts. Nordic can review it.	Antenna and radio parts already done and certified. A simpler, cheaper board is possible.
Cons	8-layer board with filled vias costs more. The ball-grid chip needs factory assembly. Bare chips are backordered (16–24 weeks in Oct 2026), so ask Nordic for samples. The antenna must be copied exactly.	Different antenna and pin layout from the DK. Limited availability (production stock arriving Oct 2026). Adds a vendor.

2.2 ECG front end: AD8232 or MAX30003

	AD8232 (chosen)	MAX30003
Pros	Simple: analog output, nothing to configure. Many tutorials. You have AD8232 boards to prototype with now. Filter corners are set by resistors and capacitors, so they are easy to change.	Digital output, so less noise pickup. 500 samples/s and 40, 100 or 150 Hz filters chosen in software. Runs from 1.8 V.
Cons	Needs a 2.0–3.5 V supply, so the board's supply moves from 1.8 V to 2.7 V. The analog signal must be kept away from the radio and the switching regulators. The datasheet's ECG circuit filters at 40 Hz, narrower than the ECGs the model was trained on (see 4.5). About 30 small parts around it.	Many registers to configure. Needs a 32.768 kHz clock input. Fewer tutorials.

2.3 Number of electrodes

	Pros	Cons
3 electrodes (chosen; datasheet Fig. 66)	A third, actively driven electrode cancels mains hum. Tells you which electrode came off.	One more electrode on the person.
2 electrodes (datasheet Fig. 62)	Fewer parts and electrodes.	More noise. Can only tell that an electrode is off, not which one.

2.4 Battery

	Pros	Cons
Rechargeable LiPo (chosen)	The nPM1300 copied from the DK already has a charger, battery gauge and temperature check. Charges over USB-C.	Needs a cell with built-in protection, and a thermistor. Never record while USB is plugged in (see Section 5).
Coin cell or AA	No charger needed.	Doesn't use the DK's power design. Voltage sags under radio and NPU load. No USB charging.

2.5 Supply voltage

	Pros	Cons
2.7 V (chosen)	Inside the AD8232's 2.0–3.5 V range. Fine for the nRF54 (1.7–3.6 V) and flash (1.65–3.6 V). The AD8232 and the chip's ADC share one supply, so the ECG signal can never go above the ADC's 0 V–VDD input range.	Slightly more power than 1.8 V.
1.8 V (DK default)	Lowest power.	Below the AD8232's 2.0 V minimum.

3. Changes to the DK design

Designators below are the DK's (schematic PCA10184 v0.7.0). Sheet numbers refer to its schematic PDF: 1 top level, 2 nRF54LM20, 3 debugger, 4 level shifters, 5 misc (LEDs, buttons, flash), 6 power supply, 7 QSPI switches, 8 connectors, 9 mechanical.

3.1 Keep unchanged

- **Sheet 2, all of it:** the nRF54LM20B and every part around it, plus their layout. That includes the antenna trace and the copper-free area at the board edge.
- **Sheet 6, nPM1300 block:** U3 with its capacitors, both inductors (L6, L7) and the LED power output (C51, C52).
- **Sheet 8, "nRF54LM20 USB" block:** J3, D1, L8, FB3, C63–C68.
- **Sheet 5:** flash U4 and C58; LED circuits LD1–LD3 (R25–R27, Q4–Q6, R29, R30, R32); button SW1; reset button SW6 with R35.
- **Sheet 8, P11 "Debug IN":** fit the header. It is left empty on the DK.

3.2 Delete

What	Designators	Why
Debugger (nRF5340) and everything on its sheet	All of sheet 3, e.g. U2, X3	You program through the SWD header instead
Level shifters and SWD switches	All of sheet 4: U5–U7, U8, U12–U14	Only connect the debugger
Flash switches	All of sheet 7: U15, U16, C61, C62, R56	Flash connects directly
Debugger USB port	J4, D2, L9, FB4, C69–C74	One USB port is enough
Power-source switch	Q2, Q3, SW5, R20–R24, C55	Only one USB port
VDD supply buffer	U9, Q8, R52–R55, C59, C60	Nordic's note on the DK schematic: not needed for the nRF54LM20 reference design
Current-measurement header, external 5 V input	P14, JP1, R15, P16	Replaced by a 0 Ω link (3.3, item 3)
Pin headers, trace header, Debug OUT	P1–P10, P12, P13, P17, P18	Development only
Fourth LED and extra buttons	LD4, Q7, R28, R31, SW2–SW4	Not needed
NFC antenna connector (optional)	J5, C56, C57, R33, R34	Only if you don't want NFC

3.3 Rewire

1. **Supply voltage:** change R17 (VSET1) from 47 kΩ to **330 kΩ**. The nPM1300 then starts its main output (VDD:nRF) at 2.7 V instead of 1.8 V. In Nordic's table, 47 kΩ gives 1.8 V, 150 kΩ gives 2.5 V, and 250–500 kΩ gives 2.7 V, which is the highest start-up setting.
2. **One supply net:** merge VDD:IO into VDD:nRF (SB20 becomes a trace; delete SB21). Leave the second output (VOUT2, 1.8 V set by R18 = 4.7 kΩ) unconnected except for a test point, as a spare 1.8 V supply.
3. **Current measurement:** replace P14, JP1 and SB19 with one 0 Ω resistor between VDD:nRF_PMIC and VDD:nRF. Removing it lets you measure current.
4. **Power chip control:** connect nPM1300 SCL and SDA (pins 14 and 13) to two nRF54 pins, with 4.7 kΩ pull-ups to VDD:nRF. On the DK the debugger drives these; on your board the firmware does, using the nRF Connect SDK's nPM1300 driver.
5. **nPM1300 VDDIO** (pin 12): connect to VDD:nRF instead of VDD:DBGR. Keep C54 (100 nF).
6. **nPM1300 GPIO1** (LED_PWR_CTRL, switches LED power on): connect to an nRF54 pin instead of the debugger.

7. **USB power:** J3 VBUS → FB3 → VBUS:nRF, which then feeds (a) the nRF54 VBUS pin through R1 (2.2 Ω), as on the DK, and (b) the nPM1300 VBUS pins through R14 (0 Ω), replacing the power-switch output.
8. **USB CC pins:** connect J3 CC1 and CC2 to nPM1300 CC1 and CC2 (pins 23 and 24), the way the DK wires its debugger port, and delete R45 and R46 (5.1 kΩ). The nPM1300 reads the CC pins to know how much current the charger can supply. Confirm with Nordic (Section 7).
9. **Flash:** wire U4 straight to the pins the DK firmware already uses: CS# P2.05, SCLK P2.01, SIO0 P2.02, SIO1 P2.04, SIO2 P2.03, SIO3 P2.00. VCC to VDD:nRF (SB24 becomes a trace; delete SB23).
10. **32.768 kHz crystal X2:** connect directly to XL1/XL2 (SB1 and SB2 become traces; delete SB3 and SB4).
11. **Reset:** R35 pull-up to VDD:nRF instead of VDD:DBGR. SW6 goes straight to the nRF54 nRESET pin, replacing SB22, SB25 and SB26.
12. **SWD header P11:** SWDIO, SWDCLK, RESET, VDD (reference, from VDD:nRF) and GND. Program the board with a 10-pin 1.27 mm cable from the DK's Debug OUT connector.

3.4 Add

1. **Battery:** 1-cell LiPo with a built-in protection circuit, on a JST-PH 2-pin connector (KiCad built-in footprint *Connector_JST:JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical*). Connect it to nPM1300 VBAT (pin 19) with a 2.2 µF capacitor to ground. 2.2 µF is the value in Nordic's nPM1300 reference designs. Pick capacity to fit the case (100–500 mAh), and set the charge current in firmware to at most the capacity per hour (for example, 200 mA for a 200 mAh cell).
2. **Thermistor:** 10 kΩ NTC (β ≈ 3380) on nPM1300 NTC (pin 18), touching the battery. Many LiPo packs have a 3-wire connector with this built in. The nPM1300 supports 10 kΩ, 47 kΩ and 100 kΩ thermistors.
3. **AD8232 front end:** see Section 4.5.
4. **Electrode connectors:** three, for LA, RA and RL. Use snap studs for adhesive electrodes, or a 3-pin connector for a lead cable.
5. **Test points:** VBAT, VSYS, VDD:nRF, VOUT2, AD8232 OUT, GND.
6. **Debug log pads:** P0.06 (TX) and P0.07 (RX), as on the DK. Logs can also go over the SWD cable (RTT) or the chip's own USB port.

3.5 Suggested pin map

nRF54 pin	Connects to	Same as the DK?
P1.22, P1.25, P1.27	LED green, amber, red (DK LED0, LED1, LED2)	Yes
P1.26	Record / upload button (DK Button 0)	Yes
P2.00–P2.05	8 MB flash	Yes
P0.06, P0.07	Debug log pads (UART0 TX, RX)	Yes
P1.03 (AIN7)	AD8232 OUT (analog input)	New
P1.11, P1.14, P1.15	AD8232 SDN, LO+, LO−	New
P1.12, P1.13	nPM1300 SDA, SCL. SCL must be on a clock pin (datasheet table 79); P1.13 is one	New
P1.10	nPM1300 GPIO1 (LED power)	New
P1.24 (optional)	nPM1300 interrupt output (e.g. charger plugged in)	New

These are suggestions. Pin choices get checked when the firmware board file is written: the SDK build rejects a pin that a peripheral can't use.

4. Supporting parts for each chip

For parts copied from the DK, the values and part numbers come from the DK's parts list (PCA10184_BOM_Web.xls). Where the DK leaves the manufacturer as "N.A.", any part with the listed value, rating and size works. For new parts, the values come from the part's datasheet.

4.1 nRF54LM20B: copy DK sheet 2 exactly

Ref	Value	Part number	Where it goes
U1	nRF54LM20B	Nordic nRF54LM20B-PAAA (98-ball FCCSP)	The chip
C4 / C5	10 µF / 100 nF	0402 X6S / 0201 X7R	VDD pins A3, A4, A9
C1, C2	100 nF	0201 X7R	VDD pins G10, H1
L1	4.7 µH	TDK MLZ1608M4R7WT000	DC/DC: DCC to DECD
C6	4.7 µF	Murata GRM035R60J475ME15J	DECD
FB1	120 Ω bead	Murata BLM03AG121SN1D	DECD/DECA supply filter
FB2, L5, C8	15 nH, 1.0 nH, 1.0 pF	Murata LQP03TN15NJ02D, LQP03TN1N0C02D, GJM0335C1E1R0WB01	DECA to DECRF filter
C7, C14	2.2 µF	Murata GRM033D80E225ME47D	DECA, DECRF
R1	2.2 Ω	0603, 1%	USB 5 V to the chip's VBUS pin
C11	4.7 µF	Murata GRM155R61A475MEAAD	VBUS pin
C10	2.2 µF	Murata GRM033R61A225KE47D	DECUSB (USB regulator)
R2	200 Ω, 1%	0201	TXRTUNE to ground. Required: it calibrates the USB lines
X1	32 MHz	Kyocera CX2016DB32000D0WZRC1 (8 pF, ±40 ppm)	XC1, XC2. Radio clock; uses the chip's internal load capacitors
X2	32.768 kHz	TXC 9HT11-32.768KDZC-T (9 pF, ±20 ppm)	XL1/XL2 (P1.20, P1.21). Low-power timekeeping
L2, C3, L3, C12, L4	3.6 nH, 1.2 pF, 4.7 nH, 1.5 pF, 3.3 nH	Murata LQP03HQ3N6B02, GJM0335C1H1R2WB01D, LQP03HQ4N7H02, GJM0335C1E1R5WB01, LQP03HQ3N3B02	Antenna tuning, from the ANT pin
C13	0.3 pF	Murata GRM0335C1HR30BA01	Not fitted on the DK; keep the footprint
J1, C76	test connector, 1.0 pF	Murata MM8130-2600; 0201 RF capacitor	Between the tuning parts and the antenna
Antenna	PCB trace	—	Copy the trace and the copper-free edge area exactly

4.2 nPM1300 power chip: copy the DK sheet 6 power block

Ref	Value	Part number	Where it goes
U3	nPM1300	Nordic nPM1300-QEAA (QFN-32)	Charger and supplies
R14	0 Ω	0402	USB 5 V into VBUS
C34, C36 / C35, C37	4.7 µF / 100 nF	Murata GRM155R61A475MEAA / 0201	VBUS
C39; C40, C41, C44, C45; C42, C43	1.0 µF; 10 µF; 1.0 nF	0402; Murata GRM188D71A106MA73D; 0201	VBUSOUT, VSYS, PVDD
R16, C38	0 Ω, 4.7 µF	0402; Murata GRM155R61A475MEAA	VSYS to the 5 V/battery rail (P5V0)
L7, C46, C47	2.2 µH, 10 µF, 100 nF	Samsung CIGT201610EH2R2MNE; GRM188D71A106MA73D; 0201	Main output VOUT1 = VDD:nRF
R17	**330 kΩ** (DK: 47 kΩ)	0201, 1%	VSET1: start-up voltage 2.7 V
L6, C48, C49	2.2 µH, 10 µF, 100 nF	As above	Spare output VOUT2
R18	4.7 kΩ	0201, 1%	VSET2: VOUT2 = 1.8 V
C51, C52	10 µF	0603 X5R	LED power output (load switch 2)
C54	100 nF	0201	VDDIO, now from VDD:nRF
New	2.2 µF	0402 X5R	VBAT, next to the battery connector
New	10 kΩ NTC	β ≈ 3380	NTC pin, at the battery
New	4.7 kΩ × 2	0201	SDA and SCL pull-ups to VDD:nRF

Layout: route PVSS1 and PVSS2 to the ground layer with vias right at the pins, as the DK does.

4.3 USB-C: copy the DK "nRF54LM20 USB" block

Ref	Value	Part number	Where it goes
J3	USB-C 2.0, 16-pin	Bellwether 80259-1021	Connector
C65, C66	10 nF, 100 nF	0201	VBUS at the connector
FB3	120 Ω, 3 A bead	Murata BLM18KG121TN1D	VBUS to VBUS:nRF
C64, C63	1.0 µF, 100 nF	Murata GRT155R61C105KE01D; 0201	VBUS:nRF
D1	ESD protection	Nexperia PRTR5V0U2F,115	D+ and D−
L8	Common-mode choke	Murata DLM0NSM900HY2D	D+ and D−, between D1 and the nRF54
C67, C68	10 nF	0402	Connector shell to ground
R45, R46	Delete	—	CC1, CC2 now go to the nPM1300 (3.3, item 8)

Layout: route D+ and D− as a 90 Ω differential pair, as the DK does on its impedance-controlled top layer.

4.4 MX25R6435F flash: copy DK U4

Flash pin	Connects to	Notes
1 CS#	P2.05	Supply range 1.65–3.6 V; runs at 2.7 V
2 SO/SIO1	P2.04	
3 WP#/SIO2	P2.03	
4 GND	Ground	
5 SI/SIO0	P2.02	
6 SCLK	P2.01	DK note: max input capacitance 6 pF, so keep traces short
7 RESET#/SIO3	P2.00	
8 VCC	VDD:nRF	C58, 100 nF, at the pin

4.5 AD8232 ECG front end: new, from the datasheet's Fig. 66 (cardiac monitor)

Copy Fig. 66 of the AD8232 datasheet (Rev. A, page 25). It is the datasheet's circuit for recording the ECG waveform: 3 electrodes, a two-pole 0.5 Hz high-pass filter, a two-pole 40 Hz low-pass filter, and a total gain of 1100. Package: 20-lead 4 × 4 mm LFCSP (CP-20-10); exposed pad to ground.

Function	Pins	Parts (Fig. 66 values)	Notes
Supply	+VS (17), GND (16), pad	0.1 µF at the pin; 1 µF nearby	+VS from VDD:nRF (2.7 V); range 2.0–3.5 V; 170 µA typical
Mid-supply reference	REFIN (18), REFOUT (8)	10 MΩ to +VS, 10 MΩ to GND, 0.1 µF to GND	Sets the signal's zero level at about 1.35 V
Electrode inputs	+IN (2), −IN (3)	Series resistor from each electrode; 10 MΩ from each input to +VS	Fig. 66 uses 180 kΩ. The datasheet's safety rule (fault current under 10 µA: R = supply ÷ 10 µA = 270 kΩ at 2.7 V) means **use 330 kΩ**. The 10 MΩ resistors are for lead-off detection
Right-leg drive (3rd electrode)	RLD (5), RLDFB (4)	360 kΩ from RLD to the RL electrode; 1 nF between RLDFB and RLD	The datasheet asks for more than 330 kΩ at 3.0 V; 360 kΩ meets that
High-pass filter, 0.5 Hz	HPDRIVE (1), HPSENSE (20), IAOUT (19), SW (6)	2 × 0.33 µF, 2 × 10 MΩ, 1.4 MΩ	Removes electrode drift and DC offset
Low-pass filter, 40 Hz, and gain of 11	OPAMP+ (7), OPAMP− (9), OUT (10)	2 × 1 MΩ, 10 nF, 1.5 nF (filter); 1 MΩ and 100 kΩ (gain)	Gain = 1 + 1 MΩ ÷ 100 kΩ = 11; total gain 100 × 11 = 1100
Mode pins	FR (15), AC/DC (14)	FR to +VS; AC/DC to GND	Fast restore on; DC lead-off detection, which 3-electrode use requires
To the nRF54	OUT (10), SDN (13), LOD+ (12), LOD− (11)	OUT to an analog input; the others to GPIOs	SDN low = off (40 nA typical); LOD+/LOD− high = that electrode is off

Filter choice to test. The 40 Hz low-pass in Fig. 66 smooths the ECG shape, but your model was trained on diagnostic ECGs, which usually keep about 0.05–150 Hz. Swapping two capacitors widens the filter: 10 nF to **3.3 nF** and 1.5 nF to **470 pF** moves the corner from about 41 Hz to about 128 Hz, with the same filter shape. The corner frequency scales with $1/\sqrt{(C1 \cdot C2)}$ when the resistors stay the same. The wider filter also lets 60 Hz mains hum through, which firmware then has to filter out. Order both sets of capacitors, record yourselves with each, and compare the model's outputs.

Layout: put the AD8232 and its parts in the corner farthest from the antenna and from the nPM1300's inductors (L6, L7), with a solid ground plane under them and short traces to the electrode connectors.

5. Building the board

- **Layer stack-up:** use the DK's spec file *400861_PCB_Laminate_Specification.pdf* (in Production files/SPEC) and send it to the board maker. It specifies 8 copper layers, FR-4, 1.57 mm total, ENIG finish, and vias filled and capped (IPC-4761 Type VII). The filled vias are needed because the chip's solder balls sit directly on vias.
- **Assembly:** have the board maker place all parts. The 98-ball chip and the 0201 parts can't be soldered by hand.
- **Programming:** the first time, the chip is blank, so program it through the SWD header with a 10-pin cable from the DK's Debug OUT, the same way you flash the DK now. That loads a bootloader (MCUboot) along with the app. After that, new firmware can go over the USB-C port or Bluetooth. The nRF Connect SDK supports both on the nRF54LM20B (release notes for SDK 3.3: DFU fully supported, and USB serial recovery on the nRF54LM20 DK). The bootloader takes flash space (Nordic's nRF54LM20 Dongle reserves 140 KB). Keep the SWD header: it is the rescue path if an update goes wrong.
- **Radio section:** keep the chip, its capacitors, the crystals, the tuning parts, J1 and the antenna in exactly the DK's positions. Keep the antenna at a board edge with no copper on any layer around it. Where you can, keep the ground area around it the same size. Keep the battery (its pouch is metal foil), the USB connector, screws and any metal case away from the antenna end; module makers suggest about 3 cm. On the body, the chest absorbs some signal, which shortens range, but a phone in the same room is well within reach. To check it, compare the signal strength (RSSI) of your board and the DK in the nRF Connect phone app at the same distance.
- **Chips:** the DK uses nRF54LM20B-PAAA. Distributors showed 16–24 week lead times in October 2026, so ask Nordic for samples early.
- **Safety:** this board is for testing on adult team members, not infants. Never record while USB is plugged into a computer or charger on wall power. The firmware should refuse to record while the nPM1300 reports USB power, and the 330 kΩ electrode resistors limit current if a part fails.

6. KiCad workflow

1. In KiCad 9, choose **File → Import → Non-KiCad Project** and pick *Altium Designer files/pca10184.PrjPcb* in the DK hardware folder. Save the result as a new project, for example *hardware/leopatch_rev1* in the repo.
2. Check the import:
   • **Board Setup → Physical Stackup** should show 8 copper layers matching the spec.
   • Refill copper zones (press B).
   • Run the design rules check (DRC) and electrical rules check (ERC).
   • Make sure connections between schematic sheets came through. The DK uses Altium "harnesses", which may not import cleanly. The board layout keeps every connection even if the schematic needs fixing.
3. Delete, rewire and add the parts listed in Section 3.
4. Add the AD8232. Download its symbol and footprint in KiCad format from Analog Devices' AD8232 product page (Symbols, Footprints & 3D Models) or from SnapEDA. The battery connector, test points and other connectors are in KiCad's built-in library.
5. Once the project is in the repo, Claude can run the rules checks, export the parts list, and write the firmware board file for the new pin map.

7. Questions for Nordic

1. We're copying the nRF54LM20 DK (v0.7.0) onto a smaller board. Is the radio section fine if the board outline and ground area around the antenna shrink? Is there a minimum?
2. Can we get nRF54LM20B-PAAA samples? Distributors show 16–24 week lead times.
3. With a single USB-C port, is it right to connect CC1/CC2 to the nPM1300 the way the DK's debugger port does, and remove R45/R46?

4. Are there newer revisions of the DK hardware files (we have 0.7.0) or the nRF54LM20B reference layout (0.6)?
5. Would you review our schematic and layout before we order boards?
6. The nRF Connect SDK includes an nRF54LM20 Dongle board (no debugger, programmed over USB, RGB LEDs). Is it available, and can we get its hardware files as a second reference?

8. Sources

- Nordic, nRF54LM20 DK hardware files v0.7.0 (PCA10184): schematic and layout PDF, parts list, laminate specification. nordicsemi.com/Products/Development-hardware/nRF54LM20-DK, Hardware files tab.
- Nordic, nRF54LM20A and nRF54LM20B datasheet v1.0: reference circuitry (10.3), USB pins and TXRTUNE (8.27), crystal oscillators, SAADC input range, pin assignments.
- Nordic, nPM1300 EK user guide: VSET resistor table. docs.nordicsemi.com/r/bundle/ug_npm1300_ek
- Nordic DevZone, "nPM1300 capacitor values: datasheet vs EK": 2.2 µF on VBAT.
- Zephyr / nRF Connect SDK v3.4.1, nPM13xx charger binding: supported thermistors (10 kΩ, 47 kΩ, 100 kΩ). nPM1300 driver for firmware control.
- Analog Devices, AD8232 datasheet Rev. A: supply range, Fig. 66 cardiac monitor circuit, input protection (10 µA rule), right-leg drive resistor, lead-off detection, decoupling.
- Macronix MX25R6435F, as used and wired on the DK (sheet 5, U4; firmware board file nrf54lm20dk).

Values marked "changed" or "new" are design decisions for this board and still need checking against the full datasheets and Nordic's review.